7. Text Generation Using Google Gemini LLM API

Develop a Python program using the Gemini API to generate responses for different user prompts and display the generated output.

Aim

To develop a Python program that connects to the Google Gemini API, accepts different user prompts repeatedly, and displays the generated responses.

Algorithm

1. Import the Google Gen AI SDK and read the API key using hidden input.

2. Create a Gemini API client with the key.

3. Accept a prompt from the user inside a loop.

4. If the prompt is exit, end the program. If it is empty, request another prompt.

5. Send each valid prompt using client.models.generate_content().

6. Display the text returned in response.text.

7. Handle API or connection errors and continue accepting prompts.

Python Program

```
from getpass import getpass
from google import genai
from google.genai import errors

api_key = getpass("Enter Gemini API key: ").strip()
if not api_key:
    raise SystemExit("An API key is required.")

with genai.Client(api_key=api_key) as client:
    print("Type 'exit' to stop.")
    while True:
        prompt = input("\nEnter your prompt: ").strip()
        if prompt.lower() == "exit":
            print("Program ended.")
            break
        if not prompt:
            print("Please enter a non-empty prompt.")
            continue
        try:
            response = client.models.generate_content(
                model="gemini-3.8-flash",
                contents=prompt
            )
            print("Generated Output:")
            print(response.text or "No text returned.")
        except errors.APIError as error:
            print("API error:", error.code)
            print("Check key, model access and quota.")
        except Exception as error:
            print("Request failed:", type(error).__name__)
```

Input

Enter Gemini API key: [hidden input]
Prompt 1: Define AI in one sentence.
Prompt 2: List two uses of AI.
Final input: exit

Output

Sample Output

```
[ ]  # Gemini text generation cell output
```

```
Enter Gemini API key: [hidden]
Type 'exit' to stop.

Enter your prompt: Define AI in one sentence.
Generated Output:
Artificial intelligence enables computers to perform tasks that
usually require human intelligence.

Enter your prompt: List two uses of AI.
Generated Output:
1. Detecting objects in images.
2. Recommending products to customers.

Enter your prompt: exit
Program ended.
```

Result

A Python program was developed to generate and display responses for different user prompts using the Google Gemini API.